Allan Daniel Gramajo

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EDUCATION

University of Georgia, Athens, GA — *Expected Graduation: Dec 2027*
Electrical and Electronics Engineering Major & Computer Science Minor — GPA: 3.50

TECHNICAL SKILLS

Languages/Standards: C/C++, Python, MATLAB, Java, C#/.NET, R, Assembly (ARM), PLC Ladder Logic
Protocols: I2C, SPI, UART, RS-422, Ethernet/PoE, GigE Vision
CAD/Software: PCB Design (KiCad), LTspice, Multisim, AutoCAD Electrical, Git/GitHub
Embedded Systems: Microcontrollers, PWM, Sensor Integration, Wi-Fi Telemetry
Test & Instrumentation: Oscilloscope & Multimeter Testing, Wire Harness Fabrication, Soldering, 3D Printing, Computer Vision

WORK EXPERIENCE

Honda Development & Manufacturing of America | *Engineering Intern* — Tallapoosa, GA | Summer 2026

- Programmed Mitsubishi PLC ladder logic to trigger an upstream pneumatic cleaning sequence for a Cognex DataMan code reader, improving read rate from 96.94% to 99.64% and cutting NoReads per 1,000 parts by 88%; validation hit 675/675 reads.
- Developed a GX Works2/GOT2000 data-acquisition system for a robotic depalletizer using pulse-triggered register storage, a rolling 100-cycle history, 32-bit averaging, HMI visualization, and USB/CSV logging for cycle-time and capacity analysis.
- Reverse-engineered an eight-camera GigE Vision thermal-imaging system and legacy C#/.NET application across Ethernet/PoE, GenICam, Pleora eBUS SDK, and Kepware paths; verified live streams on six cameras and scoped modernization.
- Prototyped the architecture for a PyTorch computer-vision inspection system for clutch housings, establishing camera-acquisition and dataset/training/inference workflows for automated small metal-chip detection and green/red pass/fail validation.

RMA Body Shop | *Part-Time Mechanic* — Riverdale, GA | Aug 2024 – Present

- Fabricated and modified custom wire harnesses and interface adapters for vehicle audio/electrical integrations to OEM wiring schematics and connector pinouts, including crimp and solder terminations, splices, routing, and strain relief.
- Ran pin-to-pin continuity, impedance, and ground-path checks; maintained a shared pinout/wire-list spreadsheet the shop reused.
- Replaced major vehicle components (engines, transmissions, radiators, fuel pumps) and performed brake service and routine maintenance across 3–5+ vehicles weekly; diagnosed electrical faults and communicated repair scope to customers.

PROJECTS

Bipropellant VTOL — Senior Capstone | *Avionics & Controls* — Athens, GA | Aug 2026 – Present

- Developing a bipropellant vertical takeoff and landing (VTOL) test vehicle with thrust vector control (TVC) on a 7-person multidisciplinary team, targeting an isolated static-fire qualification of the engine assembly and avionics prototype.
- Designing in-house electrical systems and PCBs spanning embedded avionics, MCU firmware, sensor/actuator interfacing, DAQ, telemetry, ignition sequencing, propulsion control, and remote emergency shutdown for static-fire and tethered-flight testing.

UGA Rocketry — IREC | *Avionics & Embedded Systems Engineer* — Athens, GA | Aug 2026 – Present

- Developing STM32-based flight avionics and telemetry software for UGA Rocketry's IREC sounding rocket on a 6-person embedded systems team, targeting closed-loop apogee control near 10,000 ft AGL.
- Architecting and designing the in-house main flight computer, custom PCBs, and instrumentation spanning inertial/atmospheric sensor fusion, SPI/UART/RS-422 communication buses, and real-time telemetry packetization.

Electro-Optical Materials Characterization — Research | *Photonics* — Athens, GA | Aug 2026 – Present

- Synthesize layered perovskite crystals and screen growth batches by optical microscopy for twisted, helical, and ribbon morphologies; use linear-polarization microscopy to quantify anisotropy vs. analyzer/sample angle.
- Acquire I-V sweeps on individual crystals to extract resistance, hysteresis, and contact behavior; correlate with morphology.
- Developing Stokes/Mueller polarimetry (PSG/PSA, DAQ, calibration) to quantify birefringence, dichroism, and depolarization.

Mixed-Signal Electronics Tool Bench — Course Project | *Embedded & Analog Design* — Athens, GA | Jan 2026 – May 2026

- Built a Raspberry Pi 4 mixed-signal electronics tool bench on a 5-member engineering team, designing all 5 test functions in-house as component-level circuits rather than COTS modules (100 Hz–10 kHz square/sine wave generation, 500 Ω–10 kΩ resistance measurement, ±5 V voltage measurement, and ±5 V DC reference) controlled through an embedded Python UI.
- Engineered a custom 6-bit SAR ADC-based voltmeter and DC reference path and designed its in-house KiCad PCB, spanning ±5 V with 64 ADC codes and 0.625 V reference steps; validated at 7 test points from −5 V to +5 V with ≤0.25 V absolute error.

FSAE Electric Vehicle Team | *Powertrain Engineer* — Athens, GA | Aug 2025 – Dec 2026

- Designed a manual-reset SR latching block for the Shutdown Circuit using AutoCAD Electrical, two-relay logic, inline fusing, and flyback suppression to provide fail-safe open behavior after monitor faults.
- Coordinated shutdown-circuit and low-voltage architecture with a 7-person tractive safety team and a 16-member powertrain team, establishing a single testable wiring standard and EV-tech procedure.